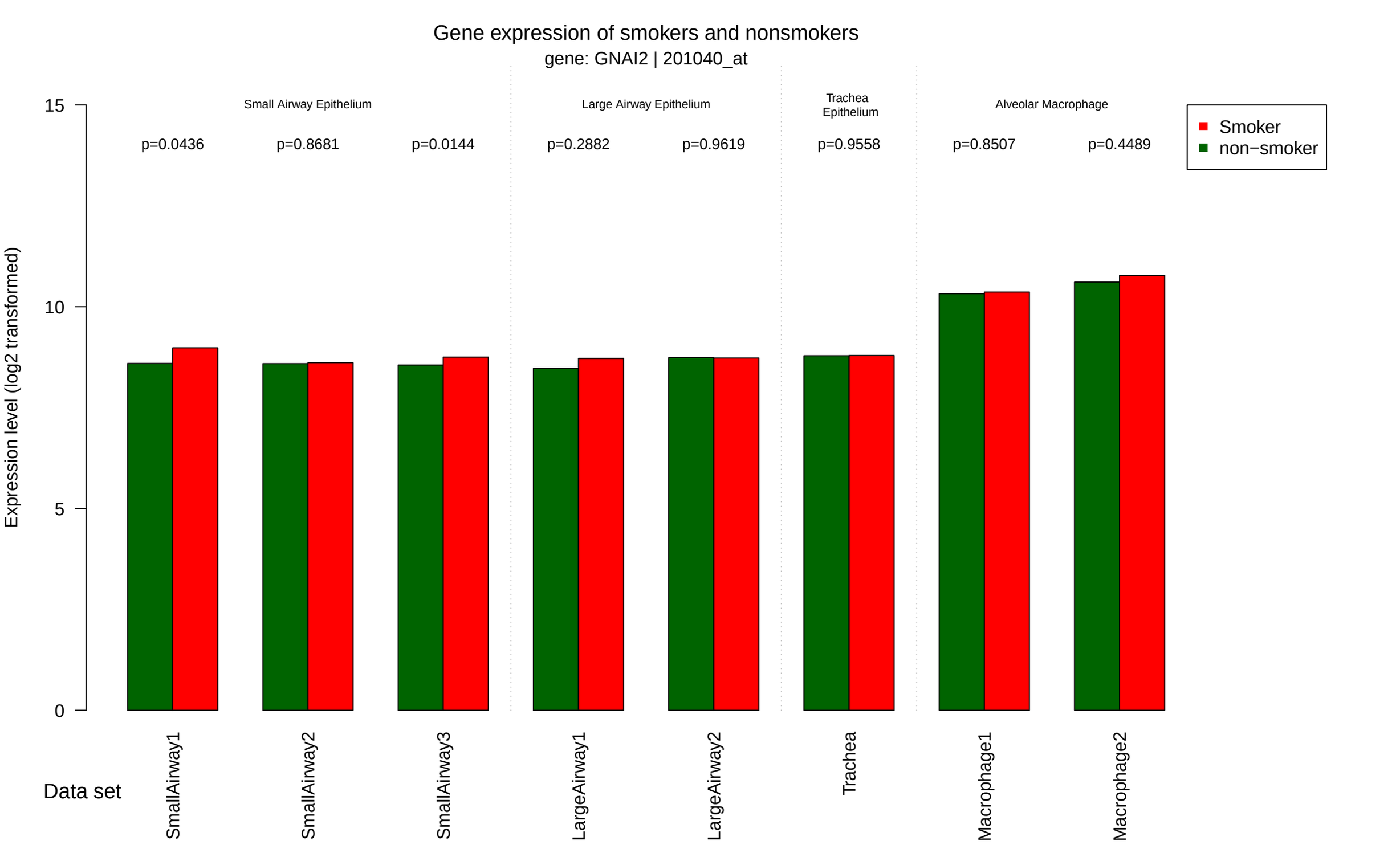
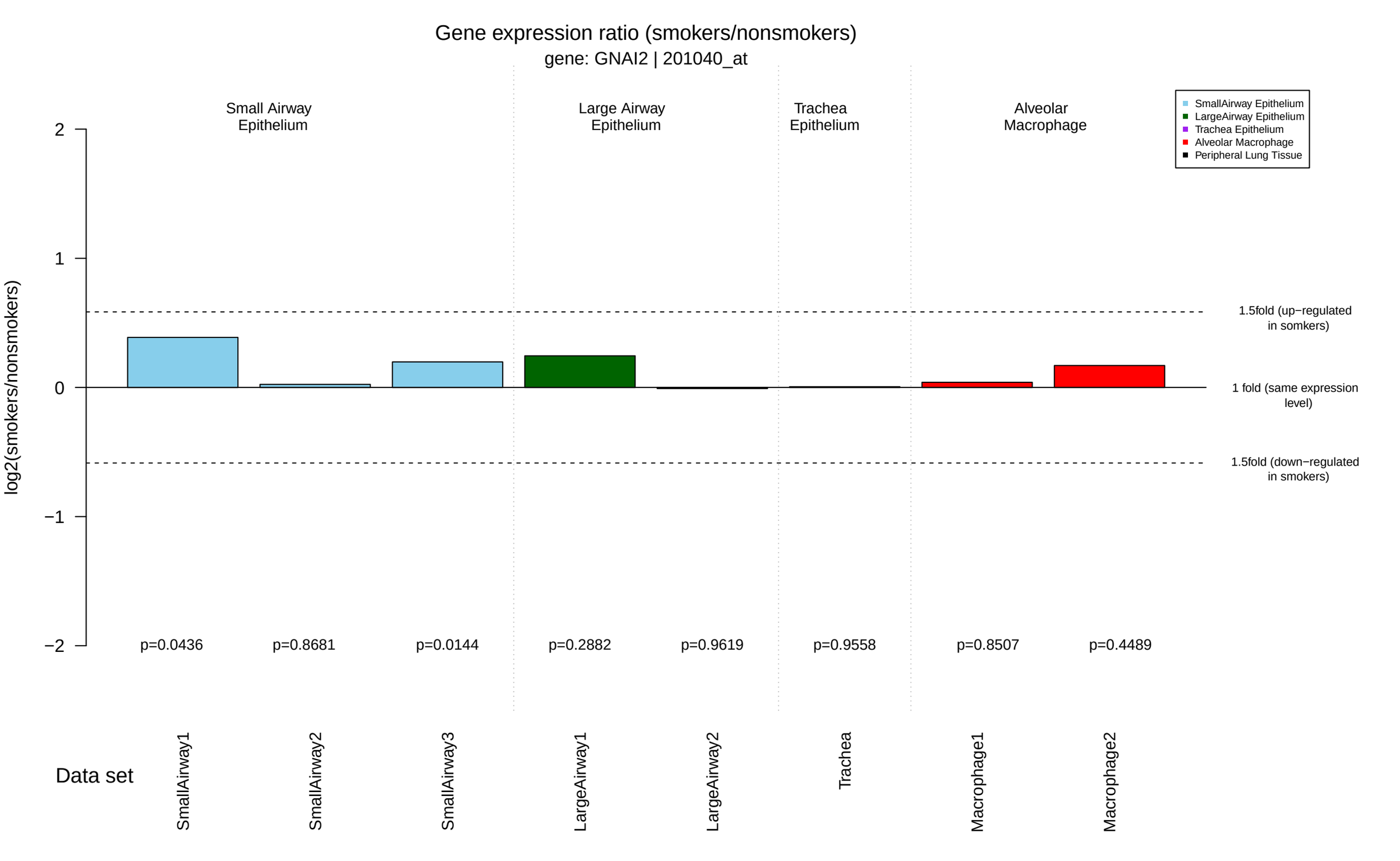
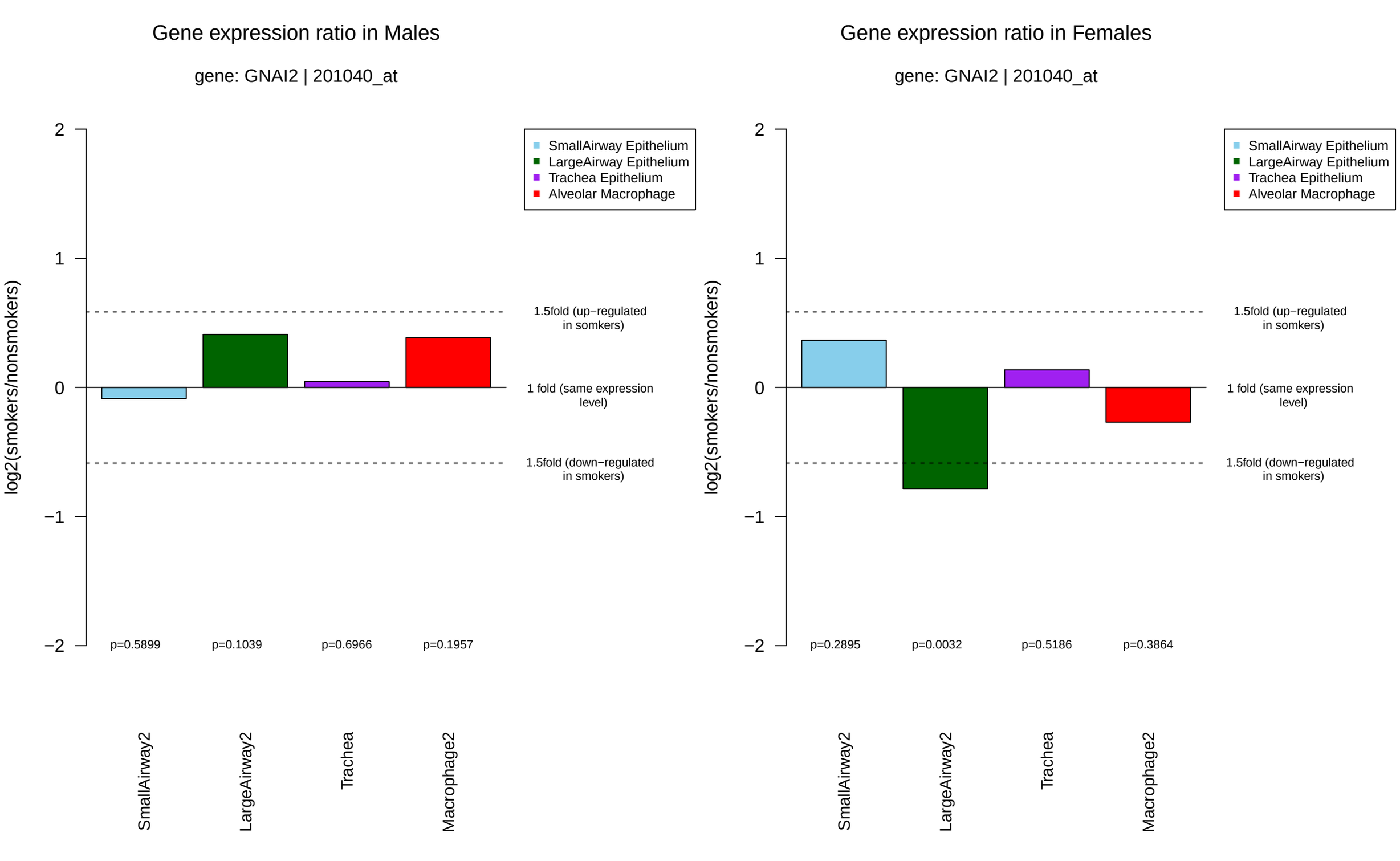
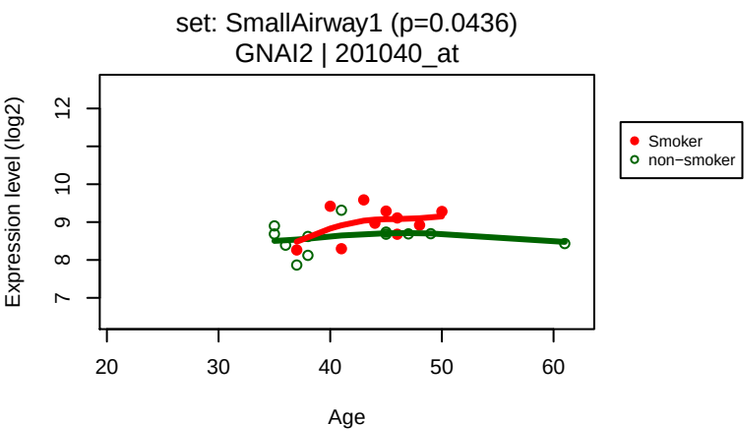
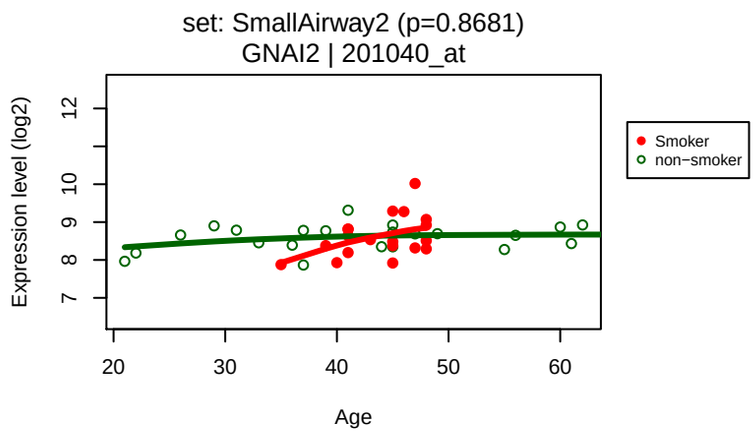
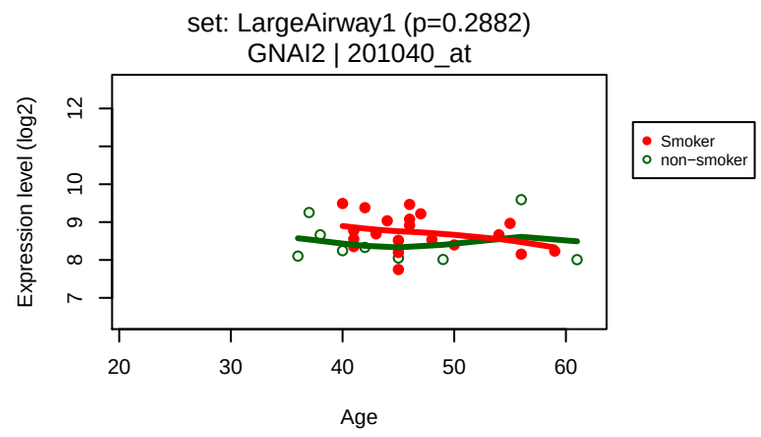

Gene expression of smokers and nonsmokers
gene: GNAI2 | 201040_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0436
p=0.8681
p=0.0144
p=0.2882
p=0.9619
p=0.9558
p=0.8507
p=0.4489
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

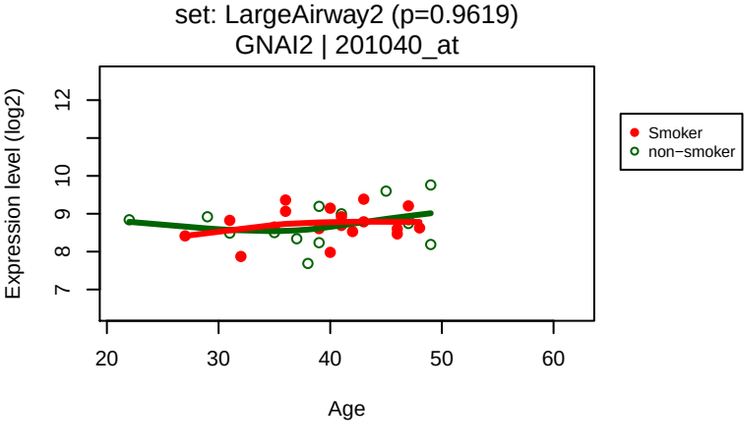

Gene expression ratio (smokers/nonsmokers)
gene: GNAI2 | 201040_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0436
p=0.8681
p=0.0144
p=0.2882
p=0.9619
p=0.9558
p=0.8507
p=0.4489
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

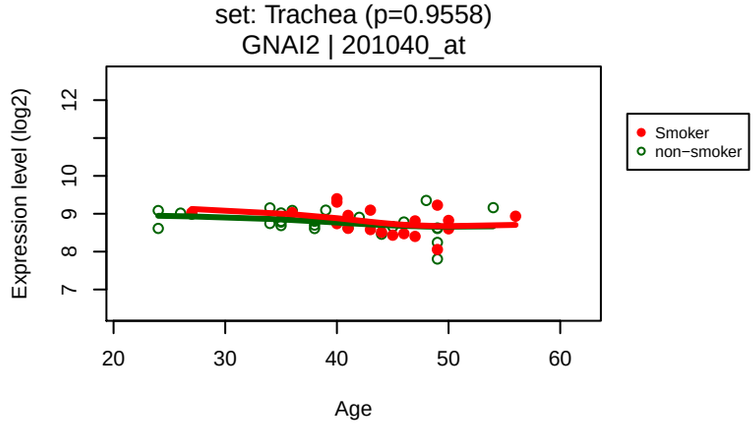

Gene expression ratio in Males
gene: GNAI2 | 201040_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5899
p=0.1039
p=0.6966
p=0.1957
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GNAI2 | 201040_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2895
p=0.0032
p=0.5186
p=0.3864
SmallAirway2
LargeAirway2
Trachea
Macrophage2

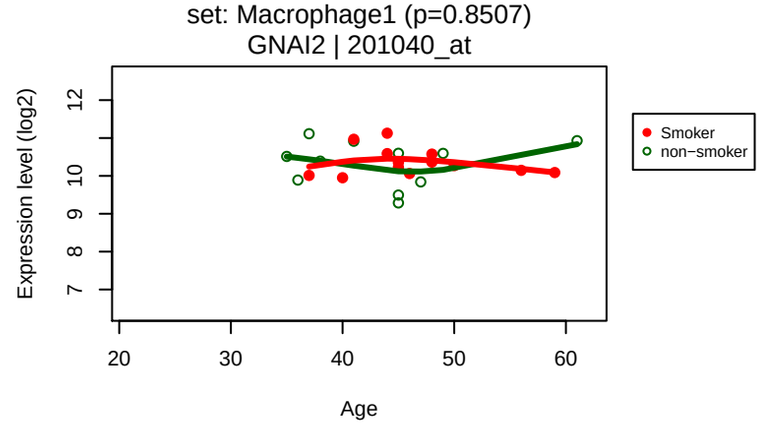

set: SmallAirway1 (p=0.0436)
GNAI2 | 201040_at
Expression level (log2)
Age
Smoker
non−smoker

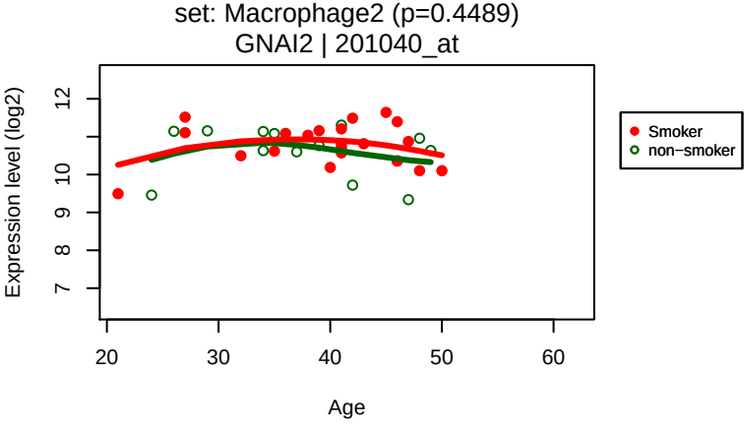

set: SmallAirway2 (p=0.8681)
GNAI2 | 201040_at
Expression level (log2)
Age
Smoker
non−smoker

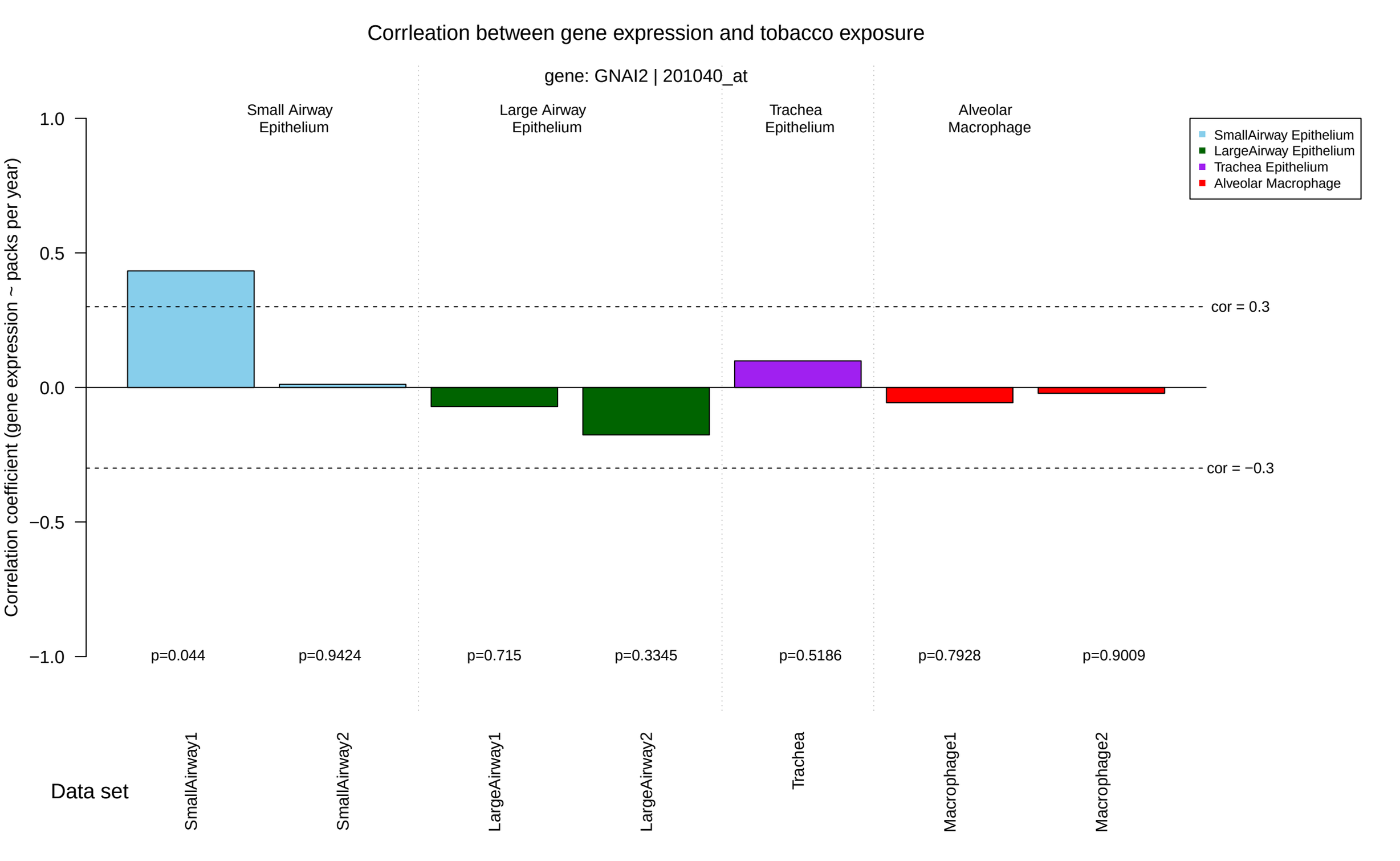

set: LargeAirway1 (p=0.2882)
GNAI2 | 201040_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9619)
GNAI2 | 201040_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9558)
GNAI2 | 201040_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8507)
GNAI2 | 201040_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4489)
GNAI2 | 201040_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GNAI2 | 201040_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.044
p=0.9424
p=0.715
p=0.3345
p=0.5186
p=0.7928
p=0.9009
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2